Pakistan has a critical water shortage as a result of the low rainfall and 30–40% water loss through leaking pipelines leading to less water available per person. This is worsened by the lack of awareness of the water consumption amongst the inhabitants and the fact that pipelines are old and families are forced to purchase water from water tankers, which is a substantial amount of their income. Diseases such as waterborne diseases can be caused by dirty water. Present reactive techniques for addressing these leakages from the pipes have been ineffective because the leakages have been undetected for a longer duration leading to more loss of water. Flowise is always surveying pipelines and strives to save water and promote good water use. The things that we need to do is to minimize NRW to the minimum level and promote responsible water use that can help mitigate the water crisis.

We suggest an internet of things (IoT)–based machine learning method for leakage detection in real time of a municipal water network. Edge microcontrollers can be connected to continuous hydraulic flow and pressure sensors, for real time hydraulic telemetry. The data is safely synchronized in the cloud-based central architecture, and scale invariant features are fed into a cloud-based, advanced hybrid deep neural network 1D CNN and Gated Recurrent Units (GRU) which is used to distinguish between physical disturbances and structural failures. To confirm the same, there are multi-channel stabilization gates which immediately alert an administrative mobile application dashboard. Finally, this system will offer a high fidelity, actionable mitigation system to the utility companies and citizens to better manage water resources.

Water crisis is a huge problem in Pakistan and worsens every day. It has fallen to almost 1,000 cubic meters per capita of water available each year, compared with 1951, when it was around 5,000 cubic meters per capita, which is the water scarcity limit for the country [3]. This is a significant reduction for several key reasons, including the growing population and climate change and dated infrastructure. There is high NRW in major urban cities such as Karachi and Lahore as a result of undetected leakage and ruptures in municipal pipelines, where loss of water supply is in the range of 30%-40% of the total water supply [6].

In the traditional infrastructure management, leaks in the physical pipe can only be found and mended after a significant amount of time has passed, which requires a lot of time and manpower resources. The seriousness of this crisis has been highlighted by recent events, including the major pipeline leak in Karachi in April 2025 which led to the loss of 400 million gallons of water [7]. The existing leak detection in municipalities is not sustainable as the leak can only be detected by humans manually, for days or even weeks.

The Flowise project suggests an automatic, real-time water management and monitoring by telemetry system to overcome these infrastructure weaknesses. Finally, Flowise aims to reduce Non Revenue Water, accelerate anomaly detection and dynamically facilitate water resource management by water authorities and end-users by implementing an Internet of Things (IoT) and Machine Learning (ML) system.

The primary motivation of development of Flowise is the need to upgrade the existing water distribution system with the use of smart, automated technologies to meet that need. A high risk of a high level of water loss driven by the traditional reactive maintenance approach will lead to a high financial cost for the water provider and the need for expensive private water tankers by citizens in the event of a water loss. By monitoring water at the edge of the network with closed-loop water valves and providing water telemetry that continuously monitors water on the network in sub-second increments, Flowise hopes to decrease physical water loss, stabilize the municipal supply lines, and save the time and trouble of manual surveys.

Furthermore, the aim of the research is to foster sustainability in the international level. Flowise's embedded IoT nodes and deep learning sequence models enable prompt support for the United Nations Sustainable Development Goals (SDGs) by efficiently managing resources and building resilient urban infrastructure, such as for SDG 6 (Clean Water and Sanitation) and SDG 11 (Sustainable Cities and Communities).